Javier Aldana
ECE 308
4/16/2026
Assignment 4

Question 1.

$$T_{ry}(s) = \frac{K\left(\frac{b_0}{s+a_0}\right)\left(\frac{s+z}{s(s+P)}\right)}{1+K\left(\frac{b_0}{s+s_0}\right)\left(\frac{s+z}{s(s+p)}\right)} * \frac{s(s+a_0)(s+p)}{s(s+a_0)(s+p)}$$

$$= \frac{Kb_0(s+z)}{s(s+a_0)(s+p)+kb_0(s+z)}$$

Question 2.

$\Delta(s) = s(s+a_0)(s+p) + Kb_0(s+z)$

➔ $\Delta(s) = d(s) + Kn(s)$

$d(s) = s(s+a_0)(s+p)$
$d(s) = s^3 + s^2(a_0+p) + spa_0$

$n(s) = b_0(s+z)$

Question 3.

$$d(s) = s^3 + s^2(160 + 60) + s(160 * 60) = s^3 + 200s^2 + 7200s$$

$$n(s) = 60s + 1800$$

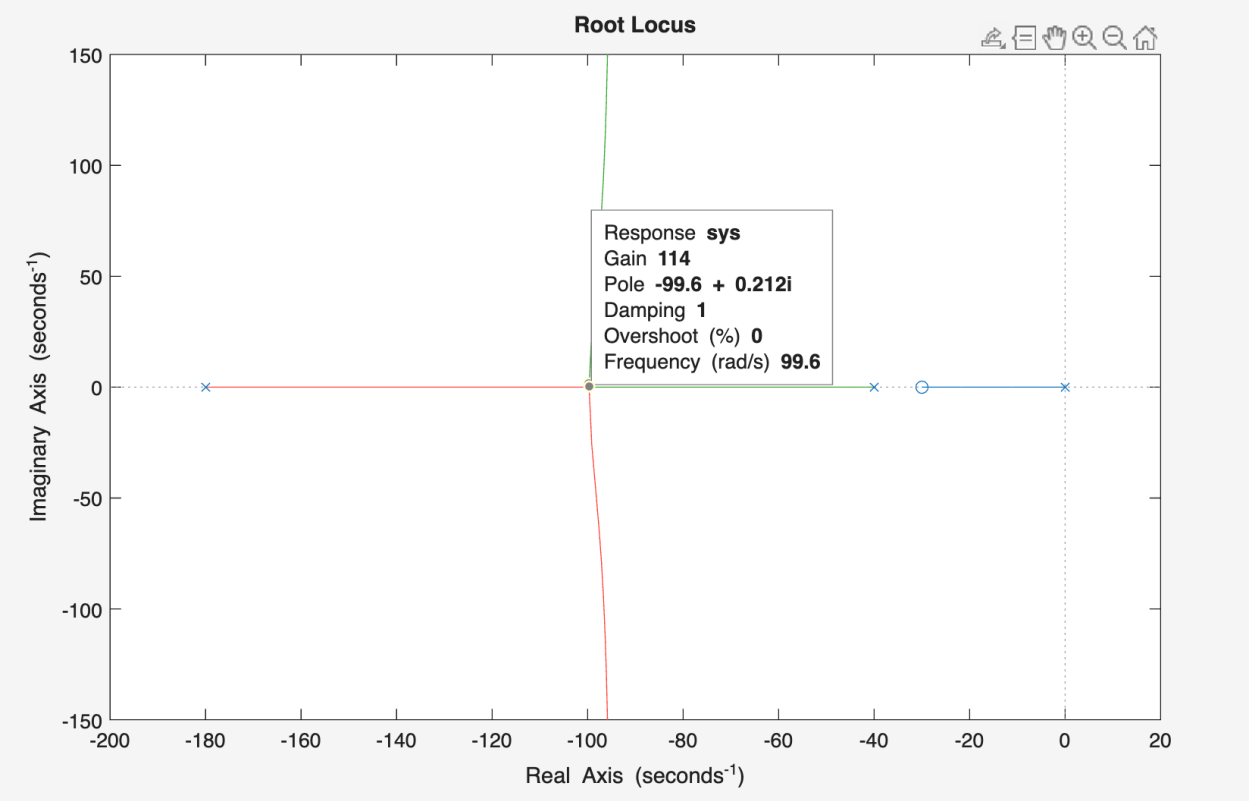


Figure 1: MATLAB Root Locus graph of $\Delta(s)$

All roots of $\Delta(s)$ will be nonnegative for K = 114

Question 4.

Final Value of K for percent overshoot of 5%: 412.

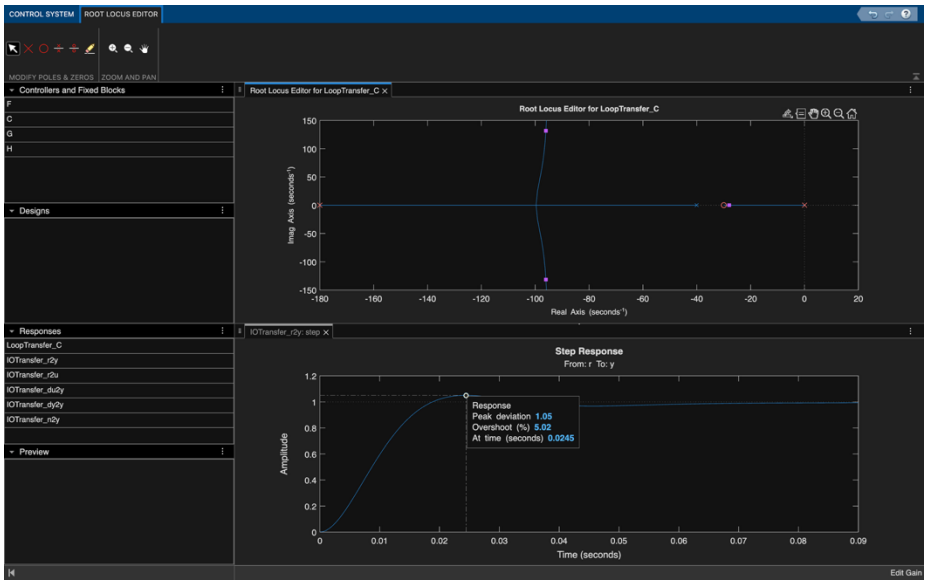


Figure 2: Final Root Locus and Step Response

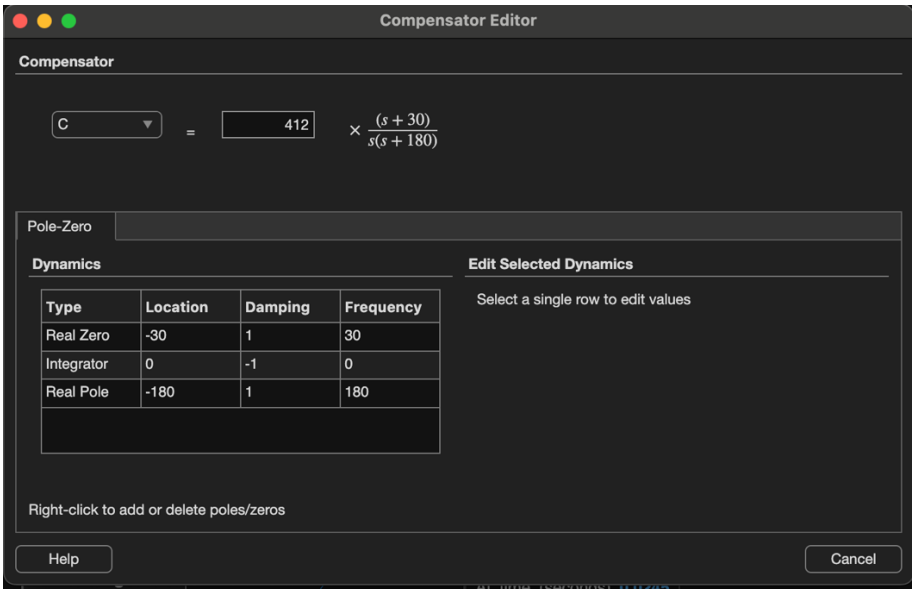


Figure 3: Control System Designer Toolbox

Question 5.

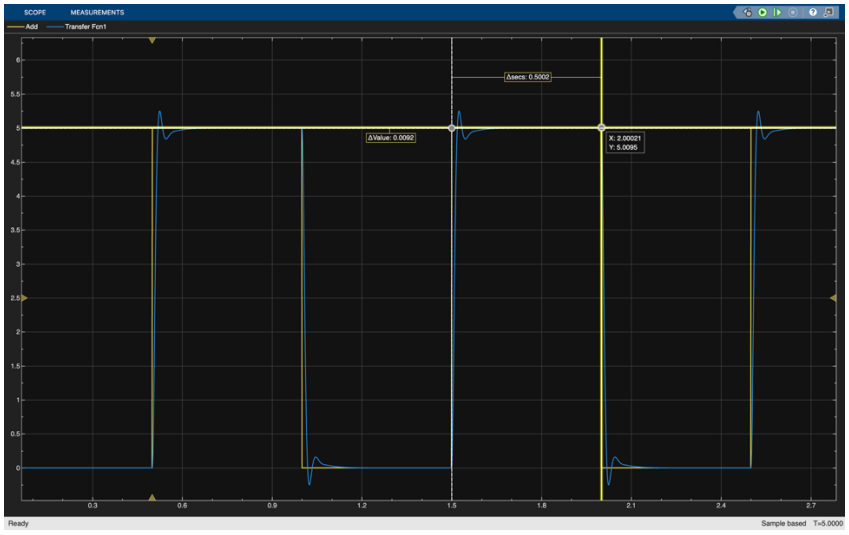


Figure 4: Reference Signal (yellow) vs Output Speed Signal (Blue)

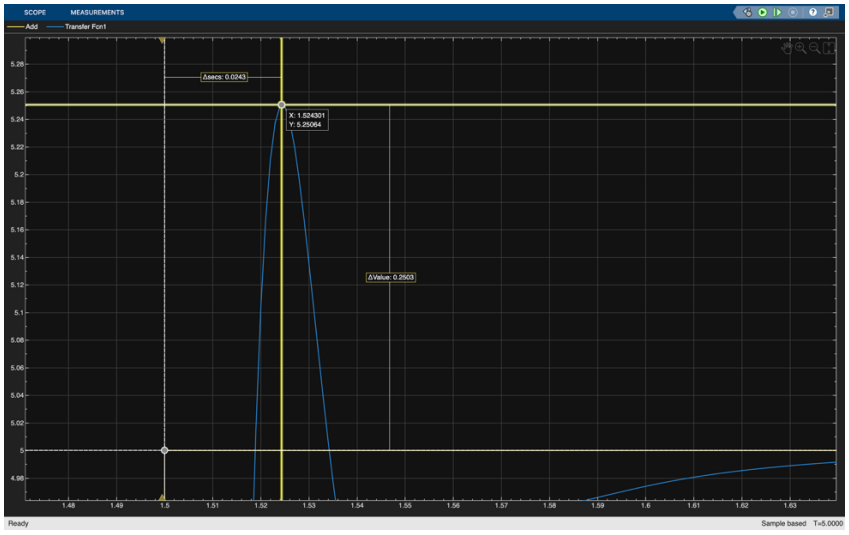


Figure 5: Zoomed In Reference Signal (yellow) vs Output Speed Signal (Blue)

Steady State Value: 5

Peak Value: 5.25

Peak Time: 0.0243

Percent Overshoot = (5.25 – 5) / 5 * 100 = 5%

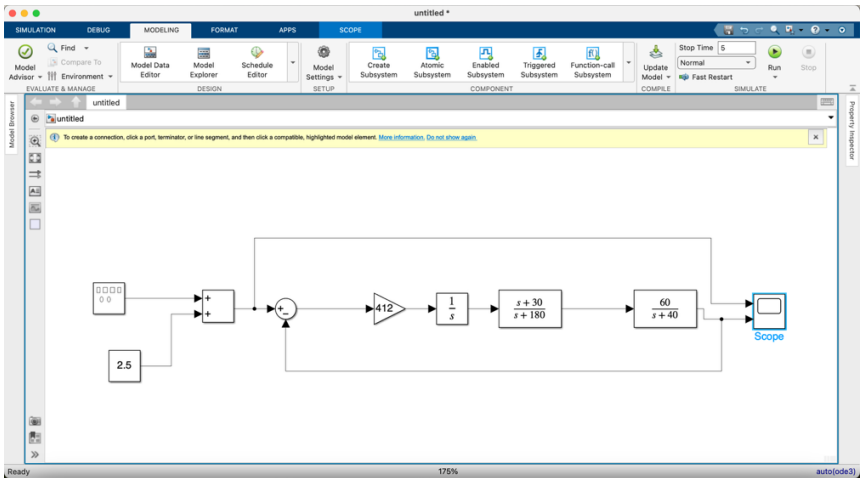


Figure 6: Simulink Model of Lead Controller